MILITARY SPECIFICATION

COMPRESSORS, RECIPROCATING, MOTOR DRIVEN, NAVAL SHIPBOARD HIGH PRESSURE AIR SERVICE

1. SCOPE

1.1 Scope. - This specification covers electric motor driven high pressure air compressors (5000 pounds per square inch gage (p. s. i. g.), maximum) for installation aboard Naval ships.

1.2 Classification. - High pressure air compressors shall be of the following classes, as specified (see 6. 1):

Class A - 20 cubic feet per hour of compressed air at 3000 p.s. i.g. (approximately 60 cubic feet per minute (c. f. m.) of standard air 1/).

Class AA - 13 cubic feet per hour of compressed air at 4500 p. s. i.g. (approximately 60 c. f. m. of standard air 1/).

Class B - 7 cubic feet per hour of compressed air at 3000 p. s. i.g. (approximately 24 c. f. m. of standard air 1/).

Class BB - 4.5 cubic feet per hour of compressed air at 4500 p.s. i. g. (approximately 24 c. f. m. of standard air 1/).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitations for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-763 - Steel Bars, Shapes, and Forgings-Corrosion Resisting

GGG-P-781 - Puller and Puller Kit, Mechanical, and Mechanical Puller Attachment.

MILITARY

MIL-R- 196 - Repair Parts for Internal Combustion Engines, Packaging of.

MIL-B-857 - Bolts, Nuts, and Studs.

MIL-S-901 - Shockproof Equipment, Class HI (High-Impact), Shipboard Application, Tests For.

MIL-D-963 - Drawings, Electrical, Hull and Mechanical Equipment for Naval Shipboard Use.

MIL-F-1183 - Fittings, Tube, Cast Bronze, Silver-Brazing.

MIL-C-2174 - Controllers, Direct-Current, Naval Shipboard.

MIL-C-2212 - Controllers, Alternating-Current, Naval Shipboard.

MIL-L-9000 - Lubricating Oil; Internal Combustion Engine, Diesel.

MIL-Q-9858 - Quality Control System Requirements.

MIL-P-15024 - Plates, Identification ,_ Information and Marking For Identification of Electrical, Electronic and Mechanical Equipment.

MIL-M-15071 - Manuals, Equipment and Systems.

MIL-P-15137 - Provisioning Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).

MIL-T-15377 - Temperature Monitor Equipment, Naval Shipboard.

1/ Standard air is defined as air at 14.7 pounds per square inch absolute (p. s. i. a.) pressure, 68° Fahrenheit (F.) temperature, 36 percent relative humidity (density 0. 0750 pounds per cubic foot (PCF).

FSC 4310

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MIL-C-15730 - Coolers, Fluid, Industrial, Naval Shipboard: Lubri - eating Oil, Hydraulic Oil, and Fresh Water.
MIL-M-17060 - Motors, Alternating Current, Integral Horsepower (Shipboard Use).
MIL-P-17286 - Propulsion and Auxiliary Steam Turbines and Gears (Including Repair Parts, Tools, Accessories and Instruments); Packaging of.
MIL-L-17331 - Lubricating Oil, Steam Turbine (Noncorrosive).
MIL-M- 17413 - Motors, Direct Current, Integral H. P. , Naval Shipboard.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-B-17931 - Bearings, Ball, Annular, Specially Selected [or Quiet Operation (Noise Tested).
MIL-A-18001 - Anodes, Corrosion Preventive, Zinc: Plate, Disc and Rod Shaped.
MIL-G-18997 - Gages, Pressure, Dial Indicating, Bourdon Tube.
MIL-H-19457 - Hydraulic Fluid, Fire Re - sistant.
MI L-F-20042 - Flanges, Pipe, Bronze (Silver Brazing).
MIL-V-20065 - Valve, Relief, Naval Shipboard.
MIL-F-21467 - Fittings, Flareless, Fluid Connection (Shipboard use).
MIL-V-22549 - Valves, Angle, Relief, Naval Shipboard for Gas Service (Except Oxygen) (Sizes 2-inches IPS and Below)
MIL-P-22808 - Paint, Epoxy, Hydraulic Fluid Resistant.
MIL-E-22843 - Equipment, Low Noise Level, (Naval Shipboard Use).

STANDARDS

MILITARY

MIL-STD-278 - Welding and Allied Processes for Machinery for Ships of the United States Navy.
MIL-STD-740 - Noise Measurements of Shipboard Machinery and Equipment.

DRAWINGS

BUREAU OF SHIPS

810-1385850 - Piping. Gage. For all Service.
810-1385884 - Unions, Fittings and Adapters Butt and Socket Welding 6000 PSI WOG and OXY (IPS).
5000 -S4800-3000 - Schedule for Piping, Pipe Fittings, Valves and Types of Joints Used in Piping Systems,
5000 -S1112-1385778 - Mount, Resilient, EES Type.

PUBLICATIONS

BUREAU OF SHIPS

NAVSHIPS 250-637-2 - Instructions for Torch Brazing of Ferrous and Non-Ferrous Piping.
NAVSHIPS 250-648-8 - Inspection and Test of Silver-Brazed Piping Systems.
NAVSHIPS 250-660-30 - A Guide for Design of Shock Resistant Naval Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connec - tion with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

COMPRESSED AIR AND GAS INSTITUTE
ASME Power Test Code (PTC9--1954 - Compressed Air Handbook, 2nd edition.

(Application for copies should be addressed to the Compressed Air and Gas Institute, 90 West Street, New York 6, New York).

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

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OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N.Y.)

3. REQUIREMENTS

3.1 Basic design drawing approval. - Air compressors furnished under this specification shall be in accordance with the supplier's class C basic design drawings approved by the Bureau of Ships prior to award of the contract or order for the class or, if more than one, the classes respectively of air compressors specified in the contract or order (see 3.7.2.3, 6.1, 6.2 and 6.3) provided that in the event of any conflict or inconsistency between the said drawings and the requirements of this specification, the latter shall govern except to the extent provided in 3.7.2.3.2.

3.2 Description. -

3. 2.1 Each compressor shall be multistage, reciprocating, vertical or horizontal design, with an intercooler and moisture separator after each stage of compression. Compressors of all classes shall be satisfactory for lubrication by either phosphate ester lubricant or petroleum oil without modification.

3. 2.2 Each compressor shall be driven by an electric motor, either directly or through a suitable reduction gear. Special attention shall be given to the design of reduction gears to insure satisfactory and quiet operation.

3. 2.3 A complete compressor unit shall consist of the compressor and motor mounted on a common bedplate, and shall include all necessary appurtenances except that motor controller, pressure switch and suction silencer may be separately mounted. Safety features, including guards for all exposed moving parts, shall be incorporated in the design. When required by the contract or order to be sound isolation mounted, a mounting arrangement shall be supplied in accordance with 3.3.6.2.

3. 2.4 All classes of compressors under this specification shall have identical parts insofar as possible, in order that parts may be interchangeable between the several classes to the maximum possible extent.

3.3 General design. -

3. 3.1 The principle of reliability is paramount and no compromise of this principle shall be made with any other basic requirements of design. It is the intention of this specification to obtain air compressers of such design that they will operate over a long period of years with a minimum of servicing. Where wear or corrosion is unavoidable the parts subjected to such wear or corrosion shall be of the best materials available for the purpose in order to reduce these detrimental effects of a minimum. The design and construction of all compressors shall be the most compact consistent with the following requirements:

(a) Reliability.
(b) Accessibility for repair.
(c) Resistance to wear or corrosion.
(d) Economy.
(e) Satisfactory operation when inclined as follows:
  (1) Up to 30 degrees in any direction (permanently inclined.)
  (2) With the ship rolling up to 60 degrees from the vertical to either side, taking 8 seconds for a complete cycle (with direction of crankshaft in fore and aft position).
  (3) With the ship pitching 10 degrees up and down from the normal horizontal plane, taking 6 seconds for a complete cycle.
(f) Quiet operation.
(g) Minimum space and weight,

3.3. 1.1 The designer should take cognizance of the conditions under which equipment will be maintained and repaired on shipboard, and of the fact that the personnel responsible for maintenance and repair may not be seasoned mechanics. For example, parts should be so designed that it is impossible to rcplace them in a wrong position;* and adjacent studs should be of the same length, or of large difference in lengths, so that a mistake will not easily occur; serrated surfaces should not be used for accurate positioning unless specifically approved by the bureau or agency concerned, since the serrations are easily damaged: large wrench areas should be avoided where strong wrenching would cause damage.

3.3. 1.2 To be in consonance with 3.3. l(g) the dry weight of compressors with 440 volt (v.) alternating current (a. c.) motors, less motor controllers, but including all other appurtenances, shall not exceed 4000 pounds. The corresponding weight with 230 v. direct current (d. c.) motors shall not exceed 5000 pounds.

3.3. 1.3 Any exceptions to the design requirements of this specification will be considered but shall be subject to approval by the Bureau of Ships. Such exceptions shall be clearly indicated at the time that the manufacturer submits basic design (class C) drawings for approval (see 3.7.2.3).

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3.3.2- Each compressor shall be capable of continuously supplying the specified quantity of air at the normal discharge pressures (see 1. 2) and standard design conditions (see 3.3. 4). All compressors shall be capable of satisfactory operation at an overload pressure of 500 pounds per square inch (p. s. i.) over their rated discharge pressure for a period of 2 hours in any 4-hour period.

3. 3.3 The temperature of air discharged from any stage of compression shall not exceed 430°F. under standard design conditions and overload discharge pressure conditions. The temperature ‘ shall be measured by temperature sensing devices (see 3.3.22.3. 1) located in the air stream as close as physically practicable to the discharge side of the discharge valves.

3. 3.4 Unless otherwise specified in the contract or order the following standard design conditions shall be assumed:

(a) Ambient and intake air temperature - 122°F.
(b) Intake air pressure - 14.7 p.s. i. a.
(c) Relative humidity -50 percent.
(d) Maximum permissible temperature of air leaving the aftercooler - 125° F.
(e) Temperature of cooling water at inlet - 95° F. for submarines; 85°F. for surface ships.
(f) Cooling medium - fresh water on submarines, and sea (salt) water on surface ships.

3. 3.4.1 Standard design conditions specified in 3. 3.4 are the conditions under which the compres - sors shall deliver their rated capacity and pressure. The compressors shall also operate satisfactorily under the following conditions encountered on shipboard, but not nessarily at rated capacity:

(a) Minimum ambient and intake temperature - 40°F.
(b) Air intake pressure - 11.7 p.s. i. a. minimum to 18. 7 p.s. i.a. maximum.
(c) Relative humidity - 100 percent at 122° F.

3.3.4. 1.1 The range of air intake pressures (see 3.3.4. l(b)) is applicable for submarine applications only, and the high intake pressure is not expected at the same time as the minimum ambient and intake temperature. The motor horsepower (hp.) rating shall be based on an intermittent duty of 2 hours with an intake pressure of 18.7 p.s.i.a. and intake temperature of 70° F. followed by continuous duty with intake pressure of 17.7 p.s.i.a. and intake temperature of 70” F.

3. 3.5 Shockproof ness. -

3. 3.5.1 The design of all complete air compressor units shall be such that they are capable of passing the type A, class HI (high-impact) shock test specified in MIL-S-901. Grade III equipment in accordance with MIL-S-901 is required. NAVSHIPS 250-660-30 may be used as a guide for design of shock resistant Naval equipment.

3. 3.5.2 Shockmounts are not to be used unless bureau or agency approval of the mount and its proposed application is given.

3. 3.5.3 Bolts designed to be stressed in shear shall be installed in holes with a minimum of clearance. Hole diameters shall be no more than 1/32 inch larger than the bolts up to 3/4 inch size, and no more than 1 ’16 inch larger than bolts of greater than 3/4 inch size. Mounting bolts for fastening the equipment shall conform to grade 2 of MIL-B-857.

3. 3.6 Noise and vibration. -

3. 3.6.1 Each complete compressor unit in - cluding the motor, shall be designed to obtain the quietest possible operation consistent with com - pressor reliability, and accessibility for maintenance and repair. Equipment shall be designed to meet the noise and vibration requirements for grade C equipment in MIL-E-22843. ‘

3.3. 6.2 When specified (see 6. 1) the manufacturer shall provide means for supporting the compressor on Navy furnished sound isolation mounts designed to Drawing 5000 -S1112-1385778. The support mounting shall be on a plane through the approximate center of gravity of the complete equipment. Such support mounting shall be integral with the subbase or other compressor supporting structures. This arrangement shall be clearly delineated on the drawings. The compressor manufacturer will not be required to furnish any mount - ing members between the isolation mounts and the ship’s structure,

3. 3.7 Gage board temperature indicating and alarm instrument, drain valves and all features of an operational nature shall be located on and operable from one side, designated as the operating side of the compressor, so as to be accessible if the compressor is installed adjacent to a bulkhead. Means shall be provided for the installing activity to relocate, to the opposite side of the compressor, the gage board with its instruments, and the drain valves, with only minor rearrangements of wiring and tubing, in order to facilitate either “right hand” or “left hand” installation. This provision shall be clearly delineated on the drawings.

3. 3.8 The crankcase, frame, housing, cylinders, and other fixed parts or attached component parts of the assembly shall be rigidly secured and held in alignment by dowel pins, fitted bolts, keys,

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rabbeted or tongue and groove Joints, or other suitable means,

3.3.9 Air cylinders. - Air cylinders shall be provided with ample water jackets. So far as practicable handholes shall be provided for access to the jacket spaces, Cylinder liners shall be provided and shall be held rigidly in place. All shoulders where piston rings may catch shall be avoided and liners shall be properly beveled. Back off or Jacking screws shall be provided to assist in disassembly at all gasketed Joints between air cylinders and between air cylinders and cylinder heads.

3.3.10 Pistons.- Pistons speeds shall not exceed 850 feet per minute (f. p. m.). Step seal design piston rings shall not be used for rings below 4 inches nominal diameter. When iron rings arc used with aluminum pistons, the aluminum shall be hardened by anodyzing to prevent excessive ring groove wear.

3.3.11 Connecting rods. - The crank end of connecting rod shal l be arranged for renewal of crankpin bearings and for such adjustment for bearing wear as may be necessary for the proper operation and maintenance of the compressor. Bolts and nuts for securing crankpin bearings shall be positively locked. Any takeup bolts and nuts shall be positively locked. Upper end of connecting rods shall be of one-piece construction.

3.3.12 Crankshaft. - The crankshaft shall be a single forging or casting, The cranks shall be arranged to give the best possible running balance. Counterweights shall be fitted if necessary. The design shall be such as to permit convenient use of a portable tachometer.

3.3.13 Flywheels, - Each compressor shall be equipped with a flywheel of sufficient size to provide smooth running of the compressor and motor, Flywheels shall be keyed to the crankshaft and shall be pressed or securely clamped thereon. A separate flywheel will not be required if ample flywheel effect is otherwise provided.

3.3.14 Bearings. -

3.3.14.1 Main bearings shall be replaceable and may be of a rolling contact type in accordance with MIL-B- 17931, or a sliding contact type, as approved by the Bureau of Ships,

3.3.14.2 Crank pin bearings shall be precision type replaceable bearings. All connecting rod bearings and bushings shall be renewable.

3.3.15 Lubrication. -

3.3. 15.1 Each compressor shall satisfy the operating conditions specified herein whether lubricated in both running gear and cylinders with a petroleum lubricant or a phosphate ester lubricant. It shall be possible to convert from one type of lubricant to the other without modifying any part of the compressor, and without modification or replacement of gaskets, “O” rings or packing.

3.3.15.2 Petroleum lubricating oil shall be in accordance with MI L-L-17331 or MIL-L-9000, Phosphate ester lubricant shall be in accordance with MIL-H-19457 but having an approximate -- - viscosity of 550 Saybolt Seconds Universal (SSU) at 100” F. and 30, 000 SSU at 40° F.

3.3.15.3 An adequate and self-contained lubricating system of the forced feed type shall be provided for the crankpin bearings, gears and pinions and [or the main bearings if not antifriction type. A suitable oil sump or reservoir shall be provided, and the oil shall be drawn from the reservoir by one or more pumps which are driven from the compressor crankshaft or motor shaft, The arrangement of reservoir and pumps shall be such that oil will not be lost from the reservoir nor the pump lose suction due to roll or pitch of the ship. The reservoir shall be provided with suitable fittings for filling and draining. The drain fitting shall be arranged so that the reservoir can be completely drained when the compressor base is level. An oil level gage, marked to show' the high, low, and normal oil levels, shall be included. Gage glasses, if used, shall be renewable without draining the oil from the reservoir.

3.3.15.4 A full flow oil filter, preferably of a self-cleaning type having an overpressure bypass, shall be installed in the pump discharge line to filter out solid particles which might damage the compressor. The filters shall not pass particles larger than 40 microns. Suitable means shall be provided for regulating the pressure of the lubricant. A device actuated by oil system pressure shall be provided to shut the compressor down in event the oil pressure is not sufficient for proper compressor operation.

3.3.15.5 The temperature of lubricant in the crankcase shall not exceed 180” F. Oil coolers shall be provided if necessary to prevent overtemperature. See 3. 3.15.7 for oil heating requirements.

3.3.15.6 Air cylinders shall be lubricated by means of a mechanical forced feed lubricator mechanically driven from the compressor. The lubricator shall be provided with its own oil reservoir of sufficient capacity to supply cylinder lubrication for at least 8 hours of continuous operation at the manufacturer's specified normal delivery rate of the lubricator, but in no case of less than

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5 pints capacity above the indicated low level. The cylinder lubrication system shall include the following:

(a) Means for adjusting the delivery rate of each individual cylinder over a wide range above and below the normal operating flow rates for either phosphate ester or petroleum lubricant. The recommended maximum flow rates and range of flow rates for both lubricants shall be indicated on the drawings.
(b) A handwheel or crank for operating the lubricator manually.
(c) Visual and quantitative flow indication to each cylinder.
(d) A check valve in each delivery line at the point where the line connects to the air system. ,
(e) A check valve at each lubricator discharge connection if necessary to prevent the lubricant from draining back to the lubricator reservoir after compressor shutdown.
(f) Means for checking oil level in lubricator reservoir.

3.3.15.7 The main lubrication system reservoir and the cylinder lubricator reservoir shall each be designed to be provided with a separate thermostatically controlled electric heater to insure adequate flow of phosphate ester lubricant under low ambient temperatures. A minimum ambient temperature of 40° F. shall be assumed in the design. Heaters may be designed to operate from either a 120 v. system or from the motor supply voltage. The automatic heater controls shall be tied in with the motor controller so as to prevent starting of the compressor before lubricant temperature will insure adequate flow.

3.3.15.7.1 The electric heaters shall be provided and installed on the compressors only when specifically required (see 6. 1). The design shall be such that the heaters, thermostats and other pertinent devices may be obtained from the compressor manufacturer in kit form and satisfactorily installed by ships' forces without drilling holes or any other machine work. This shall be thoroughly delineated on drawings and in manuals. Drawings shall clearly indicate parts included in the heater kits.

3.3.15.7.2 Consideration will be given to deletion of the requirement of electric heating of the cylinder lubricator reservoir providing the compressor manufacturer can demonstrate by actual test that the compressor cylinders will be adequately lubricated with phosphate ester lubricant at 40° F. temperature in the reservoir.

3.3.16 Bedplates. - The bedplates shall be continuous under the compressor and its driving unit.. They shall be machined to receive the intermediate frames or housing, or may be integral therewith. All top and bottom bearing surfaces shall be machine finished. Bedplates shall be sufficiently rigid to permit handling, shipment, and installation of the units onboard ship, without disturbing the alignment of the assembled units; and such that the normal distortion, weaving or vibration of the supporting structures onboard ships cannot cause misalignment of compressors. See 3. 3.6.2 regarding sound isolation mounting.

3.3.17 Housing. - The housing shall consist of cast frames cast frames and forged steel tie rods, or a structural steel welded assembly which shall support the cylinders, The crankpits and the housing shall be wholly enclosed to prevent the splashing of oil. Access openings with covers shall be provided for observation and adjustment of the bearings, and where necessary breather openings fitted with screens shall be provided. The crankcase vent or breather shall be piped to the compressor suction.

3.3.18 Air valves. - The air valves shall be of an efficient and durable design and shall be readily accessible for inspection and overhaul without disturbing air or water piping.

3.3.19 Cooling system. -

3.3.19.1 General design. - The compressor shall be provided with an efficient water cooling system, including water jacketing for cylinders, intercoolers and after cooler. The temperature of air delivered from the after cooler shall not exceed 125° F. under rated operating conditions.

3.3.19.1.1 At any gasketed joint in the cooling water system where failure or leakage of a gasket or “O” ring would permit the entry of coolant into the air stream or of compressed air into the cooling system, a double seal, vented to atmosphere between the air seal and the coolant seal, shall be provided,

3.3.19.1.2 The cooling water outlet shall be provided with a water flow indicator of the "look-through” type having an element which moves with flow, to permit observation of water flow and the detection of air in the cooling system. The cooling water outlet shall also be provided with a globe type manually operated valve for controlling the rate of flow through the cooling system.

3.3.19.1.3 Provision shall be made for automatically venting entrained air from pockets or high points in the cooling motor jackets of cylinders, cylinder heads and coolers. Provision shall also be made for completely draining all parts of the circulating water system.

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3. 3.19.2 Cooling systems for submarine air compressors. -

3. 3.19.2.1 For submarine service the cooling system shall be designed for the use of fresh water supplied [rem an unpressurized! shipbuilder furnished, closed cooling water heat exchanger system. The coolIng system shall be designed for a maximum pressure of 50 p. s. i. and shall have an attached circulating water pump driven by the compressor or by Independent means, and capable of supplying adequate flow through the compressor cooling system plus an additional dynamic head of 8 p. s. i. g. at the required flow rate for external system drop through the ship's piping and heat exchanger. When the pump is independently driven it shall be arranged to automatically stop and start with the compressor,

3. 3.19.3 Cooling systems for surface ships. -

3. 3.19.3.1 For surface ship service all parts of the cooling system shall be of corrosion resisting material and shall be designed for the use of salt (sea) water supplied from the ship's flushing or fire mains at a maximum working pressure of 175 p.s. i.g. The cooling system shall include an attached valve for positive and automatic control of the cooling water so that the water flow is maintained while the compressor is running and is stopped when the compressor is not running. The water flow should stop within 10 seconds after the compressor is stopped. A solenoid type valve shall not be used.

3. 3.19.3.2 The cooling system shall be protected from electrolytic action by the insertion of replaceable zinc anodes designed to MIL-A-18001. The zinc anodes shall have as large a surface area as is practicable for maximum electrolytic protection without causing excessive turbulence. The zinc anodes shall be easily removable for inspection and replacement.

3. 3.19.3.3 The Bureau of Ships will give consideration to a closed cooling system employing liquids other than sea water, provided it can be demonstrated that there will be a saving thereby in weight or cost, or other definite advantages. In any case the ultimate heat sink will be sea water and the heat exchanger shall be furnished integral with the compressor. If an alternate design is proposed, then all departures from specification requirements shall be thoroughly described giving reasons and advantages to be expected.

3. 3.19.4 Lubricating oil coolers furnished as separate accessories shall conform to MIL-C-15730. Such coolers fabricated as an integral part of the compressor shall conform to the basic requirements of MIL-C-15730.

3. 3.19.5 Air coolers. - Inter and aftercoolers shall be of the straight-tube type, or of a type having straight tubes with a single return bend. For air pressures of 250 p. s. i. g. or higher, air shall flow through the cooler tubes and water flow over and around the tubes. For air pressures below 250 p. s. i. g. air may flow either through or around the tubes at the manufacturers option. Cooler tubes shall be roiled, welded or brazed into tube sheets or headers, except that roiled tubes shall not be used for pressures above 250 p. s. i. g. Provision shall be made for prevention of damage due to the expansion of tube nests. Stay rods as necessary shall be provided for support of tube sheets and air heads. Each cooler tube nest shall be a self-contained unit so that it may be easily removed from the compressor as a unit for inspection and cleaning. Suitable air relief valves shall be fitted on or after each air cooler. Suitable water relief valves shall be fitted at the high point of the water side of each air cooler except for first and second stage coolers. Water relief valves shall be set at 60 p. s. i. for submarines and at 185 p. s. i. for surface ships (see 3.3. 24).

3.3.20 Condensate drainage system. -

3. 3.20.1 Efficient moisture separators shall be provided after each intercooler and aftercooler for removing free moisture and oil from the air. Each separator shall be sized to require drainage not oftener than every 30 minutes under the maximum condition of 100 percent relative humidity and 122°F. temperature of intake air, and the minimum condition of 28 “F. cooling water temperature.

3.3. 20.2 Each separator shall have a drain connection at its low point leading to an automatic separator drain device mechanically or electrically timed to completely d.rain each separator in sequence from first to final stage, Drainage from the automatic drain device shall be led to an attached condensate accumulator to be stored under pressure until all stages have been drained, after which the automatic drain device shall permit the pressurized accumulator to discharge to ambient pressure through a single drain connection on the compressor. This drain connection shall be suitable for attachment, either directly or through flexible hose, to ships piping for drainage disposal. Means shall be incorporated for automatic discharge of the condensate accumulator each time the compressor stops, in order to avoid full or partial pressurization of the drain system during shut down.

3.3.20.3 Each drain line between the separator and the automatic valve shall be provided with a suitable sight flow indicator, a check valve to prevent backflow during possible malfunction, and a manually operated valve to bypass the automatic device thus permitting manual drainage direct to the drain connection. Each element of the condensate drain system, regardless of the stage, shall be

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designed to withstand the maximum operating pressure of the final stage. The sight flow indicators and the manual bypass valves shall be located at the designated operating side of the compressor in such a manner as to be readily observed and manipulated by the operator. The mounting arrangement of these items shrill also be such that they can be readily relocated on the opposite side of the compressor to provide right or left hand installation (see 3.3. 7).

3.3.20.4 The condensate accumulator shall be connected to a separate gage on the gage board. This gage shall hav e a maximum scale reading no higher than the maximum operating pressure (or the next available scale reading above) so significant pressure indicating can be obtained at the lowest stage pressure.

3.3.20.5 The quantity of air lost as the result of opcration of the automatic drain system shall be kept to a minimum, and shall not exceed 3 percent of the rated capacity of the compressor. This shall not reduce the required output of the compressors as required by 1.2.

3.3.20.6 The Bureau of Ships will give consideration to an automatic condensate drainage system differing from that described in 3. 3.20.2 to 3.3.20.4 inclusive, provided it is demonstrated by actual test that the same end result is accomplished. The test shall be witnessed by a representative of the Bureau of Ships prior to class C drawing approval of the system.

3.3.21 Unloading. - Each compressor shall be fitted with automatic unloading devices. These devices shall discharge residual air pressure to atmosphere immediately after the compressor is stopped, so that the compressor will always be un - loaded when starting.

3.3.22 Fittings. -

3. 3.22. 1 Pressure gages. - All pressure gages shall be 3 1/2 inch dial size safety type gages in accordance with MI L-G-18997, and shall be mounted on a common gage board. The gage board and its mounting brackets shall be designed so that the board may be relocated from one side of the compressor to the opposite side, or to a bulkhead. The brackets shall be readily removable from the board,

3.3.22.1.1 Pressure gages shall be provided to indicate the air-d elivery pressure from each stage of Compression, the pressure in the condensate accumulator, the pressure in the circulating water system, and the lubricating oil pressure. All gages on the gage board shall be suitably identified by the adjacent attachment of identification plates on the gage board. Each gage shall be provided with means to indicate the normal desired working pressure, and such working pressures shall be indicated on drawings and in manuals,

3.3.22.2 Gage piping, - Gage piping shall be in accordance with Drawing 810-1385850, except that flared tubing fittings and tapered pipe threads will not be permitted, and except that root valves are not required for any gage line connected to a system which will be automatically depressuriz ed when the compressor stops. Gage cocks may have test plugs in the valve body or stem in lieu of separate test plug fittings.

3.3.22.3 Temperature instrument. - A temperature indicating and alarm instrument similar to that specified in MIL-T-15377 shall be provided on each air compressor, A selector switch shall be provided to read at least twelve temperatures, and suitable circuits to automatically shutdown the compressor in the event any one of the selected temperature sensitive points (as many as five may be selected) exceeds a predetermined amount. The shutdown temperatures shall be individually adjustable over a range of 200° F. to 500° F. An alarm indicating light shall be mounted on the face of the instrument which will operate in conjunction with the shutdown feature. The Instrument shall be located with its instrument panel readily accessible on the operating side of the compressor, and its mounting arrangement shall be such that it can be readily relocated to the opposite side of the compressor to provide for right or left hand operation,

3.3.22.3.1 Temperature sensing elements shall be provided to detect the inlet and outlet temperatures at each stage of compression, the final air discharge temperature, the lubricating oil temper - ature in the reservoir, the cooling water outlet temperature, and the temperature at any other points considered by the manufacturer to be critical.

3.3.22.3, 2 The sensing elements shall be inserted directly into the air, water or oil without the use of separate wells. Elements inserted in ar streams shall be capable of withstanding 5, 000 p.s. i.g. pressure. Elements inserted in water shall be corrosion resistant. The lead wires from the sensing elements to the indicating and alarm instrument case shall be two copper wires incased in suitable Navy type cable. Lead wires shall be readily disconnected from the sensing elements without removal of the element from the compressor, and may be grouped into a harness or bank for entry into the instrument case through a multiple, bayonet type connector.

3.3.22.4 Final check valve. - The final air discharge shall be provided with a check valve which shall be sufficiently tight to permit unloading and dcpressurizing of the compressor at shutdown.

3.3.22.5 Small manually operated valves such as gage cocks, separator drain valves, and similar items shall be of the soft seat type or tapered needle type with replaceable packing. Such valves

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in cooling water systems may be stainless steel, class 316, in accordance with QQ-S-763.

3.3.23 Air cleaner-silencers. - Provisions shall be made for cleaning the inlet air and silencing the inlet air noise. These functions may be performed by a combination cleaner silencer, or by a separate device for each function. Provision shall be made for replacement of cleaner elements, The inlet connection shall be flanged for possible attachment of an external pipe to conduct clean air to the compressor from a remote location. These devices shall be as compact as practicable consistent with effective sound attenuation (see 3.3. 6) and filtration.

3.3.24 Relief valves. - Air relief valves shall be of the soft seat type in accordance with MIL-V-22549. A valve shall be installed on or after each intercooler and aftercooler, in a location where the air temperature does not exceed 180" F. Water relief valves shall be in accordance with MI L-V-20065 except that inlet and outlet connections shall be silver-brazing unions in accordance with ?vlTL-F-1183, and their discharges shall be piped to a common drain connection suitable for attachment to ships piping. The manufacturer's piping shall be sized so as not to affect the operation of the relief valves.

3.3.25 Piping. - All piping and fittings shall conform to Drawing 5000 -S4800-3000 except as modified herein. Unions or flanges shall be used for compressor water connections to external shipboard piping systems. The unions and flanges shall be in accordance with MI L-F-1183 and MIL-F-20042 respectively. When space or weight savings can be effected, terminal connections to compressor component parts may have unions partially integrated with such parts, and may utilize the brazed or welded ring and male union nut type of construction with soft copper gaskets. Tapered pipe thread connections shall not be used except for fittings such as plugs for core holes, or the dead ends of cross-drilled passages, which are intended to keep openings permanently closed. Terminals and joints of small tubing lines mounted on the compressor shall be straight thread flareless fittings conforming to MIL-F-21467 except that working pressures above 3000 p. s. i. will be permitted in the 1/4 inch outside diameter (o. d.) size only, and except that flared type fittings may be used in salt water lines. The final discharge air connection to the ship's piping shall be a union connection in accordance with Drawing 810-1385884 with socket welding tail piece. All piping shall be designed and arranged to permit maximum accessibility to compressor parts requiring periodic inspections or repair. All silver brazed piping connections shall conform to NAVSHIPS 250-637-2; for connections 1/2 inch nominal size and larger preinserted brazing rings shall be used.

3.3.26 Threaded parts. - All threaded parts shall conform to Handbook H-28. Through bolting should be used whenever possible; if not possible, then studs are preferred to tap bolts or capscrews,

3.3.27 Welding and allied processes. - Welding and allied processes shall be in accordance with MIL-STD-278. Welding of cast nodular iron parts is not acceptable,

3.3.28 Jacking device. - Suitable provision shall be made for Jacking the compressor over if it is impractical to turn the machine by hand,

3.3.29 Painting - Ferrous castings and weldments shall be cleaned by a sulfuric acid pickling process and shot blasting (not sandblasting) to remove all scale, rust and loose weld beads, After machining, ferrous castings and weldments shall be treated by a manganese phosphate rust inhibiting process (Parkerizing or equal). After assembly, all surfaces to be painted shall be thoroughly cleaned and free from oil and grease. All external nonworking ferrous and aluminum surfaces shall be painted with two coats of light gray epoxy hydraulic fluid resistant paint conforming to MIL-P-22808. Paint shall not be applied to any internal surface. When desired for convenience or appearance, contiguous non-ferrous surfaces may also be painted.

3.3.30 Identification plates. -

3.3.30.1 Identification plates shall be furnished on each compressor and shall be type A, or D in accordance with MI L-P- 15024, except that identification plates of plastic or aluminum are not acceptable. The plates shall be secured to equipment with corrosion resistant metallic fasteners.

3.3.30.2 Identification plates shall contain data as follows:

(a) Manufacturer's name.
(b) Manufacturer's model or type and size.
(c) Classification.
(d) Manufacturer's serial number,
(e) Salient design characteristics:
    (1) Capacity
    (2) Pressure in p.s.i.
    (3) Speed of shaft in revolutions per minute (r. p. m.).
    (4) Brake hp.
    (5) Working pressures of each stage,
    (6) Lubricating oil pressure
    (7) Any other data vital to the unit,
(f) Contract-number (and item number for multiple unit orders),
(g) Federal stock number.
(h) Section for inspector's stamp.

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3.3.30.3 Each driving unit and each major accessory unit shall have an identification plate in accordance with the applicable equipment specification.

3.4 Materials. -

3.4.1 For the materials of construction of major parts of the compressors covered by this specification see table I of 6.3. This table lists a number of materials which are considered adequate for certain critical applications. However, this specification is not intended to be restrictive provided proposed alternate materials will give equal or better service, and Bureau of Ships approval is obtained prior to offering equipment in response to invitations for bids. Manufacturers are urged to make use of class C drawings (basic design drawings) to obtain approval of any desired alternate materials (see 3.7.2.3).

3.4.1.1 Table I shall be followed in accordance with the following:

(a) Where material specifications are listed in table I the selected specification, or the approved alternate, shall be used and shall be indicated on the drawings.
(b) Where material specifications are not listed for specific items in table I, the material shown or its approved alternate shall be used and shall be indicated on the drawings. The material shall be of a commercial grade or specification satisfactory for the application,
(c) Material selection for items or parts not listed in table I or for items not limited or defined by requirements contained elsewhere in this specification shall be the responsibility of the manufacturer and shall be suitable for the service required. The material selected shall be shown on the drawings,
(d) Material specifications for items covered by 3. 4.1.1 (b) and (c) shall be shown on the drawings for information only; and the drawings shall indicate by note that the referenced specifications for these items will not necessarily be followed. Physical and chemical test records are not required.

3.4.2 All internal ferrous parts in the compressed air path shall be corrosion resistant material or shall be given a corrosion resistance treatment as approved by the Bureau of Ships. This treatment shall be completely described on the drawings. Cast gray iron shall not be used except for cylinder liners, pistons and piston rings.

3.4.3 All gaskets, "O" rings, packing, or other non-metallic parts coming in contact with lubricating oil, with compressed air having entrained oil or oil vapors, or with moisture condensate containing particulate or emulsified oil, shall be of a material dually compatible with both petroleum lubricants and phosphate ester type lubricants.

3.5 Electrical equipment. -

3.5, 1 Motors. - All motors shall be in accordance with MIL-M-17413 for d. c. motors, or MI L-M- 17060 for a. c. motors, and the following:

(a) Service	Navy A.
(b) Voltage and current - characteristics	as specified in the contract or order.
(c) Enclosure	Dripproof protected to 45 degrees inclination.
(d) Ambient temperature. ,	50°C.
(c) Duty.	Continuous (for exception see 3.3.4,1.1)
(f) Speed	Constant.
(g) Winding	A. c. squirrel cage induction, D. c stabilized shunt.
(h) Insulation	Class A Or B.
(i) Bearings	Ball

3.5.1.1 The hp. rating of each motor shall be in accordance with the motor ratings, but not less than 1.05 times the brake hp. of the driven compressor under rated conditions of capacity and pressure. Intermediate hp. ratings shall be considered for motors built integrally with the compressors. Compressor balancing rings are not required for motors built integrally with the compressors

3. 5.1.2 Direct current Submarine applications. - D. c. motors for submarine application shall be suitable for the varying voltage specified in the contract or order. These motors shall be compound wound and shall have adjustable varying speed characteristics.

3. 5.1. 2.1 The following duty ratings shall apply:

Voltage	Speed	Duty
142 percent of nominal	Full-field	4 hours
100 percent of nominal	Rated full-load	Continuous
80 percent of nominal	Rated full-load	30 minutes
1/ 70 percent of nominal		10 minutes

1/ The speed at 70 percent of nominal voltage, with the field rheostat setting as required for rated full-load speed at 80 percent of nominal voltage, shall be shown on the motor drawings.

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3.5.1.2.2 The hot full-field no-load speed at any voltage from 70 percent of nominal to 142 percent of nominal, shall not exceed 90 percent of the maximum safe operating speed of the compressors as established by the compressor manufacturer. These speeds shall be shown on the motor and compressor drawings.

3.5.1.2.3 Motors shall be capable of being started at all voltages within the varying voltage range specified, assuming that at the instant of starting the compressors will always be unloaded (see 3.3.21).

3.5.1.2.4 Motors shall be equipped with overspeed trips unless the conditions specified in 3.5.1.2.5 are met. Motors shall be mechanically capable of operating at a speed of 10 percent above the setting of the overspeed trip. If no overspeed trip is provided then the motors shall be mechanically capable of operating at a speed of 10 percent above the hot no-load speed at 142 percent rated voltage with a field circuit resistance equal to 110 percent of the rated resistance value of the field rheostat.

3.5.1.2.5 Consideration will be given to omission of field rheostats and overspeed trips if all the following conditions are met:

(a) At rated voltage and full field hot, the full-load speed will be the rated speed required by the compressor for rated output.
(b) Requirements of 3,5. 1.2.2 are met.
(c) At 80 percent of rated voltage, full field hot and load encountered at rated compressor discharge pressure, the speed will not be less than 75 percent of rated full load, full voltage speed. This speed should be shown on the motor drawings.

3. 5.2 Controllers and master switches. - All controllers and master switches shall be in accordance with MI L-C-2212 for a. c. equipment, or MIL-C-2174 for d. c. equipment, and the following:

Enclosure	Dripproof.
Operation	Magnetic.
Ambient temperature .	50” C.
Insulation	Class B (Class A for coils only).
Performance	(see 3.5.2. 1.)
Control function.	A. c. motor starting. D. c. motor starting (see 3.5. 1.2).
Type	A. c. across-the-line, D. c. resistor, inductive time lag acceleration.
Protection.	Overload, low voltage protection (LVP).
Duty	Same as motor (see 3.5.1).
Master switches required	(See 3.5.2.2)
Master switch enclosures	Spray tight (remote); dripproof (local).
Master Switch location	Remote or local.

3.5.2.1 Each controller shall be provided with an automatic timing device which will permit the compressor to run after starting for a sufficient period to buildup lube oil pressure, but after which the compressor will automatically stop if the lube oil failure switch on the compressor has not closed (see 3.3.1.5. 4). Terminal connections shall be provided for all other auxiliary devices (see 3.3. 15.7 and 3.3.22. 3). The terminal *connections for electric heaters (see 3.3.15. 7) shall be provided in ail cases, whether or not heaters are required by the contract or order.

3.5.2.2 Except for d. c. submarine service each controller shall be provided with remote “ON”, “OFF”, push buttons, a local two-position “MANUAL-AUTOMATIC” switch and a remote pressure switch. The push buttons shall be of the momentary contact type. With the “MANUAL-AUTOMATIC” switch in the “AUTOMATIC” position, pressing the “Oh’” push button shall prepare the circuit for operation under automatic control of the pressure switch. When the air pressure in the discharge line is below the value set on the pressure switch, the pressure switch shall close and start the compressor. The compressor shall continue to run until the pressure rises to the value required to open the pressure switch causing the controller to stop the compressor. When the pressure again drips to the low setting of the pressure switch the compressor shall start automatically. This automatic operation shall continue in effect until the "OFF" push button is pressed. With the selector switch in the “MANUAL” position, the compressor shall start when the “ON” push button is pressed, provided the pressure switch is closed, and shall continue to run until the “OFF” push button is pressed, or until the pressure switch opens. The compressor shall not then restart until the “ON” push button is again pressed.

3. 5.2.2.1 Motor overload or compressor over temperature protection shall not be by-passed by the “ON” push button., Emergency run feature shall not be provided.

3.5.2.2.2 The pressure switch shall be in accordance with MIL-C-2174 or MIL-C-2212 as applicable. The pressure switch shall be of an adjustable type, suitable for the maximum pressure specified for the compressor and with a minimum differential of not more than 6.7 percent of the compressor rating.

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3.5.2.3 Cooling system valves and unloading valves if electrically operated shall conform to all the applicable requirements for controllers as specified in 3.5.2, except that enclosures shall be watertight (or better), Valves shall be tight-seated and all internal parts in contact with water shall be non-ferrous material except where ferrous parts are necessary for the magnetic circuit, and such parts shall be suitably nickel-p]ated for protection against corrosion. Cooling system valves shall be provided with a means of manually opening the valves in case of coil failures.

3.5.2.4 Submarine direct current applications. - D.c. controllers for submarine application shall be suitable for starting and controlling the motors at all voltages within the varying voltage range specified (see 3.5.1. 2). Operation shall be semi-automat]c, with remote push button stations, one with "START" and "STOP" buttons, and one "STOP" button only. A pressure.switch shall be provided for over pres - sure shut down only.

3.5.2.4.1 A field accelerating relay shall be provided only if the motor manufacturer considers such a device essential for this application, If a field accelerating relay is provided, the voltage range over which the motor is required to start shall be fully considered in designing the relay coils.

3.5.2.4.2 Special attention shall be given to keeping weight and size to a minimum. To this end the number of accelerating steps shall be kept to a minimum. Stock sizes of controller enclosures which are larger than necessary for the equipment shall not be used.

3.6 Repair parts and tools. -

3.6.1 Onboard. - Each set of repair parts and tools required by the contract or order shall contain the parts listed in Table II, plus any add it ional parts or tools considered by the bureau or agency concerned or by the manufacturer, to be essential in successful servicing of the compressors unless otherwise specified in the contract or order the quantity of repair parts per set shall be based on one compressor per ship (that is one compressor per set of repair parts).

Table II - Onboard repair parts.

Parts as applicable to compressors furnished	Quantity of parts furnished per set of repair parts							
	Number of compressors having identical parts							
	1	2	3	4	5	6	7	8 or more
Bearings, (except for automatic drain device) complete sets	2	2	3	4	5	6	7	8
Bolts, connecting rod, nuts, washers and so forth (crank end), complete sets	1	1	2	2	3	3	4	4
Cable, temperature sensing element (length sufficient to replace three of longest leads)	1	1	1	1	1	1	1	1
Cylinder liners, each size	1	2	3	4	5	5	5	5
Cylinder heads, sets	1	1	1	1	1	1	1	1
Couplings, flexible, complete	1	1	1	1	1	1	1	1
Gages, oil level	1	1	2	2	3	3	4	4
Gages, pressure, sets	1	1	1	1	2	2	2	2
Gaskets and packing, sets	4	8	12	16	20	24	28	32
Piston rings, sets, for each stage	4	8	12	16	20	24	28	32
Pistons, each size	1	1	1	1	1	1	1	1
Pumps, oil, complete[1/]	1	1	1	1	1	1	1	1
Rods, connecting, complete sets	1	1	1	1	1	1	1	1
Springs for automatic control devices, sets	1	2	3	4	5	6	7	8
Springs, inlet and discharge valve, sets	4	8	12	16	20	24	28	32
Springs, relief valve, sets	1	2	3	4	5	6	7	8
Temperature sensing elements, each size and type	1	1	1	1	2	2	2	2
Tube nests complete, intercooler and aftercooler, sets[2/]	1	1	2	2	3	3	4	4
Tubes, intercooler and aftercooler, where easily renewable, percent of each size and type	10	10	20	20	30	30	30	30

See footnotes at end of table.

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Table II - Onboard repair parts (cont'd).

Parts as applicable to compressors furnished	Quantity of parts furnished per set of repair parts							
	Number of compressors having identical parts							
	1	2	3	4	5	6	7	8 or more
Electric heaters, each size and type	1	1	1	1	1	1	1	1
Automatic drain device parts as follows:								
(a) Drain valve wearing parts, sets	1	1	2	2	2	2	2	2
(b) Sight flow windows	4	8	16	16	16	16	16	16
(c) Check valve springs and balls	2	4	6	6	6	6	6	6
Zinc protectors, sets	2	4	6	8	10	12	14	16
Temperature device parts as follows								
(a) Transformers, each type and size	1	1	1	1	1	1	1	1
(b) Rectifiers, each type and size	2	2	2	2	2	2	2	2
(c) Test cable assembly	1	1	1	1	1	1	1	1
(d) Relays, each size and type	1	2	3	4	5	6	7	8
(e) Resistors, each size and type	1	2	3	4	5	6	7	8
(f) Potentiometer, each size and type	1	1	1	1	1	1	1	1
(g) Indicating meter	1	1	1	1	1	1	1	1
(h) Selector knob	1	1	1	1	1	1	1	1
(i) Fuses, each size and type	10	20	30	40	50	50	50	50
(j) Light bulb	2	4	6	8	10	12	14	16
Cylinder lubricator wearing parts, sets	1	2	3	4	5	5	5	5
Filter or strainer element, each size and type	1	1	1	1	2	2	2	2
Packing, rod or piston, sets	1	1	2	2	3	3	4	4
Plunger, packed each size	1	1	1	1	2	2	2	2
Pumps, water, wearing parts for, sets	1	2	3	4	4	4	4	4
Rod, piston, packed, each size	1	1	1	1	2	2	2	2
Valves, discharge, for all stages, complete with seats, guards, springs and so forth, sets	1	1	2	2	3	3	4	4
Valves, inlet for all stages, complete with seats, guards, springs, and so forth, sets	1	1	2	2	3	3	4	4
Valve discs or strips, suction and discharge, complete sets	2	3	4	5	6	6	6	6
Working parts for automatic control devices, sets	1	1	2	2	3	3	4	4
Wrist pins, sets	1	1	2	2	3	3	4	4

1/ Pump wearing parts only shall be furnished when pumps are built in such that complete assembly cannot be furnished.

2/ Complete tube nests are required only when tubes are welded or brazed.

3.6.1.1 The quantity of each particular item in a set of repair parts is based on the number of compressors having that item, even though the compressors may be of different classes (see 3.2, 4), For example, a set of repair parts may be intended to service an installation of 3 class A and 2 class B, compressors; the quantity of such parts as are used on all classes shall be based on 5 compressors; any parts used on class B compressors only shall be based on 2 compressors.

3.6.2 Repair parts for motors, controllers, pressure switches and other devices shall be furnished as required by the applicable specifications.

3.6.3 Special tools. -

3. 6.3.1 Maintenance tools. - Each set of special maintenance tools required by a contract or order shall consist of tools such as babbitting mandrels, trams for setting piston clearances, cooler tube expanders and test rings, and gear wheel, flywheel, and coupling pulling tools, required for the proper repair and maintenance of the compressors. Each set of special tools shall be furnished in a separate

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steel box. Note to ordering activities: Normally the number of sets of special maintenance tools should be the same as the number of sets of onboard repair parts; however, the number of sets may be increased as required by the size and organization of the ships.

3.6.3.2 Operating tools. - Each set of special operating tools required by a contract or order shall consist of tools, such as valve wrenching plates, normally required to operate one compressor. Each set of specia l operating tools shall be furnished in a separate steel box. Note to ordering activities Normally a set of special operating tools shall be ordered for each compartment in which an air compressor of the type ordered is installed.

3.6.3.3 Tools which are available in the Navy Stock List of General Stores, (copies of this stock list may be consulted in the office of the Government Inspector) such as common wrenches and standard pullers, will not be required. Standard complete or limited sets of pullers or parts of puller sets in accordance with G GG-P-781 shall be identified as to type and use in notes on the outline or section assembly drawi ngs, Special tools shrill be detailed, and included in the list of material.

3.6.4 Stock, - The selection, stock numbering and quantities of repair parts for stock shall be in accordance with MI L-P-15137.

3.6.4.1 Unless otherwise specified in the contract or order, the cognizant supply demand control point for repair parts for stock is Ships Parts Control Center, Mechanicsburg, Pennsylvania.

3.7 Drawings. - All drawings and lists shall conform to MI L-D-963, with exceptions and addi - tional requirements contained herein.

3.7.1 Drawings of the following classes shall be furnished as required by the contract or order:

Class A - Master drawings
Class B - Certification data, outline drawing, drawing lists.
Class C - Basic design drawings
Class D - Ship equipment drawings

3.7.2 Class A and Class D drawings .- Class A and class D drawings shall conform to approved class C drawings and shall be so complete as to permit:

(a) Evaluation of performance and quality of the equipment against requirements of this specification,
(b) Evaluation of operational and personnel safety.
(c) Procurement Of parts by the supply system.
(d) Installation, operation, maintenance and repair, without manufacturer's assistance.

3.7.2.1 Class A and class D drawings, covering air compressors under this specification, are identical as far as form and content are concerned. Class A drawings and any modifications thereof shall be approved only by the Bureau of Ships. Bureau of Ships field representatives may approve the drawings as class D drawings only (thereby authorizing manufacture without waiting for Bureau of Ships action), and at the same time or subsequently may recommend to the Bureau of Ships that the drawings be considered as class A drawings.

3.7.2.1.1 Master plans or type I drawings furnished under earlier editions of MI L-D-963 and the earner edition of this air compressor specification are considered as fulfilling the requirements of class A drawings under this specification, regardless of the source of approval of those drawings. However, approval of modifications, to those drawings after the date of this specification shall be by the Bureau of Ships only.

3.7.2.2 Class B drawings. - Class B drawings shall be furnished as a supplement to all class A and class D drawings required under a contract or order. They arw not required in connection with class C drawings.

3.7.2.3 Class C drawings, - Class C basic design draw ings covering air compressors under this specification are to be submitted to the Bureau of Ships prior to and independent of invitations for bids or requests for proposals or quotations. An approved class C drawing is a prerequisite for award of contract or order (see 6. 2). Each drawi ng may cover several classes of compressors so long as they are of the same basic design.

3.7.2.3.1 Class C drawings shall show basic design with sufficient clarity, completeness and detail as to permit a thorough evaluation of characteristics, performance and quality of the design against the requirements of this specification and to provide reasonable assurance to the Bureau of Ships that air compressors manufactured in accordance with the design will meet the requirements of this specifica - tion.

3.7.2.3.2 A class C drawing may indicate a variation in materials from that called for in Table I, and if approved by the Bureau of Ships that variation in material shall take precedence over the corresponding material in table I subject to 3.7.7.1. All such material variations shall be specifically pointed out in the transmittal letter. Approval of a class C

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drawing does not preclude the requirement for class A or D drawings and class B drawings applicable to 4 particular contract or order.

3.7.3 Format .- The manufacturer's commer - cial format may be used if the requirements for drawing content (see 3.7. 4), material identification (3.7.5), and drawing identification (see 3.7.6) are satisfied.

3.7.4 Drawing content. -

3.7.4.1 General requirements for all classes.

3. 7.4.1.1 "Manufacturer's Use Only ' notes. - Information intended for manufacturer's use only shall be so designated. Conversely) on a commer - cial drawing with wide usage it is premissible to designate portions thereon, "For Navy Use Only".

3.7.4.1.2 Dimensional tolerance.- Unless otherwise indicated by note, legend, or key it will be assumed that all tolerances shown on a drawing must be complied with during both manufacture and maintenance repair.

3.7.4.2 Type and contents of class A and class D drawings. - Drawings of the types specificed in 3. '7.4.2.1 through 3. 7 .2.3, 1 shall be furnished

3.7 4.2.1 Sectional Assembly Drawings.- Sectional assembly drawings shall include a sectional assembly and elevation drawings with complete list of material, with references to applicable detail drawings. The list of material shall include every part required in the compressor assembly, including those parts not required to be detailed. This may necessitate some side or partial views in order to show parts not otherwise shown in the main section view. The list of material shall include an indication of each part required by this specification to be furnished as an onboard repair part (this is not a repair parts list).

3.7.4.2.1.1 If more than one sectional assembly or elevation drawing is required to identify all parts, or if the sectional assembly with list of material results in a larger drawing than size 34 by 44 inches, then the list of material may be on a separate master drawing.

3. 7.4.2.1.2 Sect ional assembly drawings shall show all critical ass embly and running cleara nces The sectional assembly drawings shall also reference the preproduction test, high impact shock test, and inclination test reports, and the bureau or agency letters approving those reports, except that when a general arrangement drawing (see 3. 7. 4.3. 1) is used, this data may be referenced on the general arrangement drawing.

3.7.4.2.2 Subassemblies furnished as repair parts in lieu of or in addition to individual repair parts shall be recorded on the related outline drawings by reference notes or as additional items on the "List of Material" of the sectional assembly drawing,

3.7.4.2.3 Detail drawings.- Detail drawings shall be furnished of all parts and subassemblies necessary for evaluation of the equipment, and of all parts necessary for maintenance, and overhaul of the equipment. Details of these parts shall be so complete as to permit emergency manufacture by a Naval shipyard without assistance from the original manufacturer. Details of such parts as crankcases and cylinder blocks shall be furnished, but the dimensioned development of cored hydraulic passages need not be shown. Subassemblies whose parts can not be procured or serviced individually, should be shown as a single part and so indicated. Multidetail drawings are preferred but monodetail drawings may be used.

3.7.4.2.3.1 It is recommended that details of parts common to all or several classes of compressors be grouped together on the same sheet (s), and likewise that all details applying to a single class be grouped together on the same sheet(s).

3. 7,4.3 Contents of class B drawings. - A class B outline drawing including certif ication data shall" be furnished as a supplemental drawing to all class A and class D drawings. Outline drawings shall cover complete units, including compressor, driver, bedplate and accessories. Separate class B drawings shall be furnished under each contract or order unless the complete equipments covered by the outline drawing and the referenced drawings are in fact identical in all respects to previously approved class B drawings.

3.7.4.3.1 If the compressors and motors are close-coupled or built together so that they are not in fact separate component parts then the outline drawing may be considered as a master drawing, entitled "General Arrangement and List of Reference Drawings", and all certification data applying to an individual contract or order (including electrical controller reference drawings) should be omitted from the general arrangement drawing. In this case separate certification data sheets including all data applicable to the individual contract or order shall be furnished, and the list of repair parts may be included in the certification data sheets or on a separate master drawing to be referenced by the certification data sheets.

3.7.4.3.2 In addition to the certification data required by MI L-D-963 the class B drawings or

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"General Arrangement" drawings shall contain the following:

(a) Complete performance data of compressors, prime mover, and all auxiliaries.
(b) Table of weights of individual components and weight of complete unit,
(c) Identification of piping fittings which connect to shipbuilder's piping, showing their size and type.
(d) Center of gravity of the compressor component and of the complete assembly.
(e) Radii of gyration of complete assembly about each of the three principal axes (required only if equipment is sound isolated),

3.7.4.3.3 Onboard repair parts list. - A list of onboard repair parts shall be furnished with class B drawings, This list shall be prepared on "Provisioning List" forms in accordance with MIL-P-15137 with the additional requirement that the list shall be assigned a manufacturer's drawing number. The forms shall be assigned sheet 2, sheet 3 and so forth of the class B drawing. This list shall be furnished whether or not the parts are required in a compressor contract or order. The list shall not be modified to indicate parts or quantities of parts furnished under supplementary or separate contracts or orders. The list is intended as a record of recommendations of the manufacturer and compressor purchaser at the time of the compressor purchase. The list of onboard repair parts actually purchased shall be proposed as required by MIL-P-15137 and need not include a manufacturer's drawing number.

3.7.4.4 Contents of class C drawi ngs.- The purpose of class C drawings is specified m 3.7.2.3. These drawings shall consist of small scale (not necessarily to scale) plan and elevation views showing over-all dimensions. If the plans cover a family of compressors all of the same design, the dimensions may be tabulated. The range of capacities, pressure and speeds shall be shown. These drawings shall include an undimensioned sectional assembly with complete list of material which shall include every part required in the compressor and auxiliary assembles. Cooling systems, lubrication systems of running gear and cylinders, and automatic separator drainage systems shall be clearly delineated. If materials of some parts may change with various compressor characteristics, a table shall be in - cluded to show these variations. Alternate subassembly arrangement of different parts may be included as desired.

3.7.4.4.1 Detail drawings of individual parts are not normally required in connection with class C drawings, but may be included if desired by the manufacturer for clarification. However, detail drawings of individual parts shall be included in specific cases if required by the Bureau of Ships for evaluation and approval.

3.7.5 Material identification. -

3.7.5.1 Preferred material reference. - Where material specifications are required by 3. 4 to be indicated on the drawings, and where mater ials of iden - tical or equal quality can be identified by more than one specification or standard, the drawings need reference only one such specification or standard. In selecting the specification or standard to be referenced the following is the order of preference:

(a) Federal Specification.
(b) Military specification.
(c) Industry and Technical Society specification or standard.
(d) Manufacturer's specification or standard.

3.7.5.2 Material substitutions. - Where materials other than those covered by (a), (b), and (c) of 3. 7.5, 1, are referenced and approved, the drawings shall show the complete chemical and physical properties of the approved material. In addition, the drawings shall identify the material in terms of the Federal or Military specification, most nearly applicable, in order to enable Naval repair facilities to make emergency repair as necessary from Navy material stocks.

3.7.6 Drawing identification. - As indicated in MI L-D-963, Bureau of Ships drawing numbers are no longer assigned, hence, the manufacturer's drawing number is now the sole reference identification. "Manufacturer" in this case is defined as the company which produces the part, or parts, shown on the drawing,

3.7.6.1 Prime vendors who purchase items from subcontractors shall use the subcontractor's drawing number as the single reference identification in all cases where the part(s) delineated thereon are produced by the subcontractor. The prime vendor shall not add his drawing number to the drawing except as an unofficial reference outside the drawing border or margin.

3.7.6.2 Prime vendors who purchase parts from subcontractors for final production, test or selection phases in their own shops, have the option of using as the single drawing identification either their own title block and drawing number or the title block and drawing number of the subcontractor, but not both.

3.7.7 Prints for approval. -

3.7.7.1 Approval of drawing by either the Bureau of Ships or its field representatives does not relieve the vendor or manufacturer of complete responsibility

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for furnishing equipment meeting all performance requirements and characteristics required by the contract or order.

3.7.7.2 Prints of the following class A and class D drawings shall be submitted for approval prior to manufacture of equipment: Sectional assembly and elevation drawings (see 3.7.4.2. 1), outline drawings and detail drawings of the following parts: Bedplates and sound isolation arrangements, crankshafts, crankcase sections, cylinders, pistons, connecting rods, air valves, intercoolers, aftercoolers, separators, diagrams of all fluid systems, bearings, automatic separator drain system, temperature monitoring system, lubricators and gage board layout. (Note: If manufacture has been authorized immediately upon signing of a contract based on previously approved class C drawings, then class A or class D drawings shall be submitted for approval within 60 days after the date of the contract)

3.7.7.2.1 After approval, each drawing shall be lettered or stamped with the approval notation which shall include reference to the letter of approval and the drawing class for which approval is received (class A and class C from Bureau of Ships only). In addition, the notation on the sectional assembly drawing shall indicate satisfactory compliance with this compressor specification,

3.7.7.2.2 Responsibility for all other detail drawings. (3. 7.4.2. 3) rests with the vendor. Each such drawing shall bear a notation indicating review and approval by a responsible official of either the vendor or his subcontractor, The notation shall include the name or initials of approving of ficial, his title and the date of approval. Prints of these drawings need not be furnished to the Bureau of Ships or its field representatives prior to delivery of the final drawings, unless specifically requested.

3.7.7.3 Prints of class B drawings shall be submitted for approval at the same time as submittal of the class A or class D drawings'. If the class B drawing is submitted covering previously approved class A drawings, prints of the class A drawings shall accompany the class B drawing for informa - tion.

3.7.7.4 Prints of class C drawings shall be submitted to the Bureau of Ships for approval (see 3.1 and 3.7.2.3).

3.7.8 Drawings for drivers and associated equ ipment shall be in accordance with the specifications for those equipments.

3.8 Equipment manuals (technical manuals) (instruction books). -

3.8.1 Unless otherwise specified in the contract or order, equipment manuals shall be in accordance with requirements of MIL-M-15071. A separate manual shall be furnished for each different compressor unit except in special cases when manuals covering more than one compressor may be approved. Preliminary manuals submitted for approval shall include all proposed sections complete.

3.8.2 Illustrations. - The following illustrations shall be included m the manuals:

(a) Two photographs, taken 180 degrees apart on a horizontal plane.
(b) Reduced size reproductions of approved outline, assembly and sectional drawings of the complete unit.
(c) Approved drawing excerpts and exploded view.s of parts or assemblies requiring detailed instruction.
(d) Master drawings covering the motors and controllers.
(e) Complete operation and maintenance instructions covering electrical equipment.
(f) Thorough Instructions for changing from one type of lubricant to the other.

3.8.3 Distribution. - Two copies of manuals shall be packed with each compressor unit. Additional copies of manuals shall be furnished as specified (see 6. 1). If final manuals are not available at the time compressors are shipped then two copies of preliminary manuals shall be packed with each compressor unit; and the final manuals shall be delivered at a later time in accordance with MIL-M-15071.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for-the performance of all inspection requirements as specified herein. The government reserves the right to perform any of the Inspections set forth in the speci fication where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality control system. - The contractor shall provide and maintain a quality control system acceptable to the Government for the supplies covered by this specification. The system of quality control shall be in accordance with MIL-Q-9858 supplemented as specified in 4. 2.1 to 4. 5.4.7 inclusive.

4.2.1 Description of procedures. - Procedures shall be assembled m manual form and shall indicate organization and responsibility for the control of quality.

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4.2.2 Inspect Ion during manufacture, - The con-actor shall establish and maintain inspection at ppropriately located points in the manufacturing recess beginning with the receipt of raw materials nd subcontracted parts through final testing and reparation for shipment of completed units, to ssure continuous control of quality of parts, com - onents and assembles.

4.2.3 The contractor shall provide and maintain ages, Instruments and other measuring and testing quipment and devices necessary to assure that sup-lies conform to contract requirements. These evices shall be calibrated against measurement tandards or designated measuring equipment at stablished periods to assure continued accuracy. 'he contractor shall prepare and maintain a Written chedule for the maintenance and calibration of such quipment based on the type, purpose and degree of sage.

4.2.3.1 All instruments used in preproduction and performance tests shall be calibrated before and after tests.

4.3 Inspection of material. -

4. 3.1 Materials used in the compressors and accessories and identified on approved drawings by Federal and Military specifications listed in Table I or elsewhere in this specification shrill be subjected o inspection requirements of the applicable speci-ication by which identified.

4.3.2 Materials used in the compressors and accessories which are identified on approved draw-ings by Industry and Technical Society specifications or standards shall be subjected to inspection require - ments of the applicable specification or standard by which identified,

4.3.3 The contractor shall establish a system of inspection and identification to insure that speci-fied and approved materials are used in accordance with approved drawings.

4.3.3.1 Upon request of the Government inspec - tor, the contractor shall furnish samples of mate-rials and available information concerning their quality and use. When the identity or quality of an item is m doubt, and in the absence of valid and acceptable test data, the contractor shall conduct such tests as are necessary to determine or verify its identity or quality,

4.4 Preproduction tests. -

4.4.1 Prior to the offering of any air compres-sor for acceptance by the Government an air com-pressor of the class offered shall have successfully undergone preproduction tests as described herein-after, except as follows:

4.4.1.1 A successful preproduction test per-formed under one contract will not have to be repeated under a subsequent contract unless a deficiency in dcsign or workmanship develops with a compressor m service, or unless a modification in design is con-sidered by the Bureau of Ships to be extensive enough to warrant retest.

4.4.1.2 A successful preproduction test of a class AA compressor will be considered as covering this requirement for all other classes. Similarly a successful test of a class A compressor will\ be con-sidered as covering the requirement for a class B compressor, however a test of a class A or class B compressor will not be considered as covering class AA or class BB compressors.

4.4.1.3 All air compressors subjected to prepro-duction tests shall be sound isolation mounted even though the production compressors under the partic - ular contract are not required to be sound isolation mounted.

4.4.2 The preproduction tests shall include initial performance tests required by 4.5 followed by a run-ning test of not less than 500 hours of actual running time. The 500 hours running time shall not be con-tinuous, but shall include at least G rest periods of not less than 8 hours each,

4.4.2.1 All data required by 4. 5.3 shall be taken and recorded at the beginning and end of each running period, and at intervals of 4 hours or less during the running times.

4.4.2.2 The first 300 hours of runni ng time shall be with phosphate ester lubricant in both cylinders and running gear. Upon complet ion of this portion of the test the compressor shall be disassembled suffi-ciently to permit inspection as required by 4. 5. 3, 6. The compressor shall be thoroughly cleaned of the phosphate-ester lubricant and then filled with petro-leum oil in both cylinders and running gear.

4.4.2.3 The last 200 hours of running time shall be with the petroleum lubricant. Any changes or adjustments made at this time in the comp I essor, such as change in cylinder lubrication rate, shall be completely described in the test report. Following the completion of this portion of the test the machine shall again be disassembled sufficiently to permit inspection as required by 4.5.3.6. Changes in appearance or physical condition of parts from the previous inspection shall be noted and described in the test report. Note and comment on the accumula-tion of carbon deposits on air valves and in coolers.

4.4.2.4 If any serious wear or other deficiencies are noted, which would necessitate replacement of parts or overhaul of the compressor in less than 1000 hours of operation, these facts shall be immediately communicated to the Inspector of Naval Material and

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the Bureau of Ships. The compressor shall be left unassembled pending a decision of the Bureau of Ships.

4.4.2.5 If no serious wear or other deficiencies are noted, these facts shall also be immediately communicated to the Inspector of Naval Material and the Bureau of Ships for information. The compressor shall be cleaned up, worn parts shall be replaced, and the compressor prepared for shipment in accordance with requirements of the contract.

4.4.2.6 Within 30 days after successful comple - tion of the preproduction tests copies of a complete test report including all recorded data shall be dis - tributed as follows:

(a) Two to the Bureau of Ships (attention Code 649).
(b) Two to the purchasing activity (if other than the Bureau of Ships).
(c) One to the primary inspection of ficer.
(d) One to the branch inspection office (if any).

4.4.2, 6.1 Bureau of Ships approval of the test report shall be obtained prior to delivery of any air compressor of a class covered by the test. A period of 30 days, after receipt in the Bureau of Ships, shall be allowed for the Bureau of Ships action.

4.5 Tests of production compressors. -

4.5.1 Hydrostatic and silver brazed Joint tests. -

4.5.1.1 Hydrostatic. - The various parts of each compressor shall be hydrostatically tested in accordance with the following:

(a) Cylinders, 1-1/2 times the maximum designed working pressure.
(b) Water jackets, 265 p.s. lg. for surface ships, 100 p.s. i.g. for submarines.
(c) Intercoolers, and aftercoolers, 265 p.s. i.g. for surface ships; 100 p.s. i.g. for submarines, on the water side and 1-1/2 times the maximum design working pressure on the air side.
(a) Piping and other parts not mentioned herein and subjected to pressure shall be tested to 1-1/2 times the maximum designed working pressure.

4.5.1.2 Silver-brazed joints, - All silver-brazed joints in oil, air and salt water piping shall be tested as required by NAVSHIPS 250-648-8.

4.5.2 Capacity test methods. -

4.5.2.1 The preferred method for determining compressor capacity is by the high-pressure flask method, by noting the time required to fill a flask of known volume from zero gage pressure to the rated discharge pressure of the compressor. During the time interval required to fill the flasks the compressor discharge pressure shall be maintained constant at the rated discharge pressure. The calibration of the flask volume shall be made in the presence of the Government inspector.

4.5.2.1, 1 At the option of the compressor manufacturer or testing laboratory, the compressor capacity tests may be in accordance with the ASME Power Test Code (PTC9-1954) as outlined in the second edition of the Compressed Air Handbook.

4.5.3 Test procedures. -

4.5.3.1 Electrical .- Motors and controllers shall be tested as required by the applicable specifi - cations.

4.5.3.2 Sampling .- Completely assembled compressors, including driving units, temperature controls, unloading valves, automatic separator drain equipment and all other equipment and fittings shall be selected for testing in accordance with table III.

Table III - Sampling for performance tests.

Number of identical units on simultaneous orders	Random selection
Up to 20	1 from every 5 units
21 to 100	1 from every 10 units, but not less than 4
Above 100	1 from every 20 units, but not less than 10

4.5.3.3 Performance tests. - Compressor units selected in accordance with table III shall be submitted to a continuous running test of 8 hours duration at normal capacity and pressure. Test is to be run under normal electrical conditions specified in the contract or order. Either phosphate ester or petroleum lubrication may be used during these tests unless a specific requirement is included in the contract or order. Power consumption tests shall be conducted as a part of the tests. For these tests the data specified in 4.5.3.4 shall be recorded at the start and

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finish of the run and at regular intervals, not exceeding 1/2 hour throughout the run.

4. 5. 3.4 Recorded data. -

(a) Mechanical:
  (1) Ambient shop conditions:
    a. Dry bulb ambient temperature.
    b. Wet bulb ambient temperature,
    c. Intake temperature (in air stream close to intake).
    d, Barometer readings. ,
  (2) Air discharge pressure from each stage of compression.
  (3) Final air temperature at discharge from aftercooler.
  (4) Air temperatures before and after each stage of compression,
  (5) Circulating water relet and outlet temperatures.
  (6) Quantity of cooling water in gallons per minute.
  (7) Revolutions per minute of driving unit and of compressor.
  (8) Temperature of lube oil in crankcase.
  (9) Time required to fill flask of measured volume.
(b) Electrical:
  (1) Volts, amperes and frequency.
  (2) Power input to motor.
(c) Operational
  (1) Operation of the automatic temperature control.
  (2) Operation of the automatic unloading devices,
  (3) Operation of the automatic moisture separator drainage system.
  (4) Operation of moisture separators. (Include record of quantity of oil and water drained manually at 15-minute intervals from each separator for a period of 2 hours during preproduction tests).
  (5) Operation of lube oil heater and thermostat.
  (6) Total amount of cylinder lube oil used during test period,

4.5.3.4.1 Computed and recorded data, - The following data shall be computed from the recorded data and shall be included on each test data record:

(a) Capacity in cubic feet of air per hour actually delivered at rated discharge pressure.
(b) Brake horsepower at the compressor shaft (computed from previous test of ' motor).
(c) Volumetric efficiency.
(d) overall efficiency of compressor that is, compression efficiency versus mechanical efficiency. Theoretical air horsepower upon which efficiency is based may be calculated from either the adiabatic or isothermal base, but test data should indicate which is used.

4.5.3.4.2 During each operation test the functioning of the electrical and mechanical equipment shall be smooth and free from excessive heating, noise, and vibration and the performance m all respects shall meet the requirements of this specification.

4.5.3.5 Noise tests.- During the latter part of each performance test of compressors required to be sound isolated, airborne and structure borne noise tests shall be conducted. Testing and reporting shall be m accordance with MIL-STD-740. Details of compressor noise test stand, instrumentation and test measurements shall be as mutually agreed upon between the compressor manufacturer and the Bureau of Ships.

4.5.3.5.1 During the preproduction tests (see 4. 4) a series of four noise tests, shall be conducted as follows:

(a) During the latter part of the initial 8-hour performance test (see 4.5.3. 3).
(b) During the final hours of the 300-hour test (see 4.4.202).
(c) During the first four hours of the last 200-hours test (see 4.4.2.3).
(d) During the final hours of the 200-hours test.

4.5.3.6 Upon completion of the performance tests, each compressor and its access ories shall be disassembled sufficiently to permit inspection of all parts subject to wear or damage by running, these parts to include bearings, wrist pins, pistons and piston rings, cylinders, and air valves. To be acceptable no parts of the compressors shall show any signs of failure or excessive wear, The condition of the compressors at the conclusion of all tests shall be noted on the of ficial test sheets.

4.5.317 Each compressor unit not selected in accordance with table XII shall be subjected to tests, exactly as required by 4.5, 3.3 and 4.5, 3.5 except:

(a) The duration of the running test need nor exceed 4 hours, and data is required to be taken only at the start and finish of the run and once in between;
(b) The units need not be disassembled upon completion of the performance tests,

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unless excessive heating, noise, vibration, or erratic performance is noted during the tests.

4.5.3.7.1 If excessive heating, noise, vibration, or erratic performance is noted and is corrected, the 4-hour test shall be repeated. If the frost correction does not completely eliminate the trouble, then after further correction the compressor shall be subjected to a complete 8-hour test before being offered for acceptance.

4.5.3.7.2 All recorded data and computations required for the 8-hour test are also required for the 4-hour test.

4.5.3.8 Inclined operation. - One compressor unit. of each design, including the driving motors, shall be selected for tests to determine satisfactory compliance with 3. 3.1 (e). Inclined operation tests shall be without resil nt mounts.

4.5.3.8.1 Only one compressor of each design shall be subjected to the inclination tests, and the tests need not he repeated on later contracts or orders unless the Government inspector or the Bureau of Ships considers that sufficient changes have been made from the tested design to warrant ret esting, Neither changes in material only of compressor construction, nor changes in the driving motor shall be considered as reasons for retesting.

4.5.3.8.2 The compressor unit shall be operated at rated load for not less than 30 minutes with the base tilted 30 degrees in each of the four cardinal positions, right, left, front, and rear. During the progress of these tests there shall be no exces - sive heating of any parts, and noise and vibration shall not exceed that of normal horizontal operation.

4.5.3.9 The suitability of the compressor unit for operation when a ship is rolling or pitching shall be determined by inspection of the design,

4.5.4 High-impact shock test. - One compressor unit including driving motor, of each design, shall be shock tested in accordance with MIL-S-901 for grade III equipment. Shock tests shall be without resilient mounts.

4.5.4.1 only one compressor of each design shall be subjected to the shock tests, and the tests need not be repeated on later contracts or orders unless the government inspector or the Bureau of Sh ps concerned considers that sufficient changes have been made from the tested design to warrant retesting. Changes in materials of construction may require retesting. Changes in the driving motor only shall not. be considered a reason for retestin g the compressor however changes in the motor will require retesting of the motor.

4.5.4.2 Consideration will be given to waiving the shock test requirement on a compressor unit providing that compressor and motor is of the same basic design as that of another class which has successfully passed the shock test. The decision as to whether a shock test will be waived shall rest with the Bureau of Ships.

4.5.4.3 If a manufacturer is unable to conduct the shock tests at his own plant, he may arrange to have them conducted at a commercial or Government laboratory suitably equipped to conduct the tests. When the shock tests are to be conducted at a Government laboratory a complete set of master and outline drawings shall accompany the equipment.

4.5.4.4 The compressor shall be idle and running on alternate Mows of the shock test, but the compressor shall not be compressing air while running.

4.5.4.5 Inspection after shock test. -

4.5.4.5.1 Motor .- Inspection of the motor shall be in accordance with MIL-M-17060 or MIL-M- 17413.

4.5.4.5.2 Compressor .- Inspection of the compressor shall be conducted as follows:

(a) Examine the unit carefully and note the extent of any external damage.
(b) Jack the compressor over several times by hand to insure that all working parts are free and ready for starting.
(c) Run the compressor unloaded for a short time and note any mechanical peculiari - ties.
(d) Gradually build up the load and carefully check any undue noises or excessive vibration or other nonconformances.
(e) Secure the compressor and disassemble for complete inspection. Observe and record all effects of the shock test on the various compressor components,

4.5.4.6 Failure to pass test. -

4.5.4.6.1 Motor .- The motor will be considered to have failed the shock test unless it conforms to MIL-M-17060 or MIL-M-17413.

4.5.4.6.2 Compressor. - The compressor will be considered to have failed the shock test in the event' of the following:

(a) Breakage of any parts, including accessories and mounting bolts.
(b) Detachment of any part or object from the unit.
(c) compressor requires attention or adjustment or replacement of parts to enable

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it to perform its principal function during and after each blow of the test,

(cl) Unit fails to function satisfactorily as specifically defined in this specification.

4.5.4.7 Disposition of shock-tested equipment, - Equipment which has successfully passed the shock tests will be considered acceptable for such service as designated by the Bureau of Ships. The contractor shall thoroughly recondition such units, correc - ting any minor imperfections or deformations, and shall place the equipment in sati sfactory working order The motors shall be reconditioned and tested in accor da nce with the requirements of the motor Specifications. After reconditioning, the equipment shall be tested as spccified in 4. 5. 3.3.

4.5.4.7.1 Equipment which has been subjected to the high-impact shock tests and has failed will not be accepted, either in whole or in part until the causes of such failure have been corrected by the manufacturer, and the complete equipment has been retested, However, failure of the compressor only shall not cause rejection of an otherwise acceptable motor, nor shall failure of a motor only cause re - jection of as otherwise acceptable compressor.

4.5.5 Test reports, -

4.5.5.1 Test reports shall be prepared for all tests conducted by the contractor including the shock an d inclination tests. Recorded data of the com - pressor per formance tests may be logged on contractor's forms. The following identification data for each compressor unit shall be furnished and in - cluded in the test report:

(a) Serial numbers of the compressor and driving unit.
(b) Ship for which intended.
(c) Contract or order number.
(d) Actual net weight of the unit in the “dry” and “wet” condition.

4.5.5.2 Distribution of compressor performance test reports shall be as follows:

(a) Two to the Bureau of Ships.
(b) Two to each supervisor of shipbuilding for each ship's worth of compressors furnished (one for supervisor's files; one for shipbuilder) for privately built ships..
(c) Two to each Naval shipyard for each ship's worth of compressors for Navy built ships.
(d) One for primary inspection officer.
(e) One for branch inspection office (if any).

4.5 5.3 Distribution of shock test reports and inclination test reports shall be as follows

(a) Six to the bureau or agency concerned, for approval.
(b) One for the primary inspection office.
(c) One for the branch inspection of fice (if any).

5. PREPARATION FOR DE LIVERY

5.1 Domestic shipment and early equipment install ation and for storage of shipboard repair parts. -

5.1.1 Compressors, -

5.1.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioriation and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing - Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical or mecnanical damage during direct shipment from tile supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Class i - fication Rules and Regulations or other carrier re - gulations as applicable to the mode of transportation.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The informa - tion shall include nomenclature, Federal stock number or manufacturer's identification, contract or order number, contractor's name ant] destination.

5.1.2 Onboard repair parts. - Onboard repair parts shall be preserved and packaged level A, packed level C and marked level A and C respectively in accordance with MIL-R-196, MIL-P-17286 and MIL-E-17555 as applicable.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation, packaging, packing and marking for ship - ment shall be as speci fied by the procuring activity (see 6. 1).

(5.2.1 The following provides various levels of protection during domestic shipment anti storage or overseas shipment, which may be required when procurement is made.)

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5.2.1.1 Preservation and packaging -

5.2.1.1.1 Compressors. -

5.2.1.1.1.1 Level A.- Compressors shall be preserved and packaged in accordance with the level A requirements of MI L-C-3600.

5.2.1.1.1.2 Level C.- Preservation and packaging of compressors shall be sufficient to afford adequate protection against, corrosion, deteriora - tion and physical damage during shipment from the supply source to the using activity and until ‘early installation

5.2.1.1.2 Repair parts. - Repair parts shall be preserved and packaged in accordance with the level A requirements of MIL-R- 196, and MIL-E-17555 as applicable

5.2.1.2 Packing -

5.2.1.2.1 Compressors. -

5.2.1.2.1 1 Level A.- Compressors shall be packed for shipment in accordance with the overseas requirements of MI L-C-3600, except that plot pack tests are not required

5.2.1.2.1.2 Level B. - Compressors shall be packed for shipment in accordance with the overseas requirements of MI L-C-3600 except that pilot pack te sts are not required, and except that shipping containers shall be of the domestic type or class.

5.2.1.2.2 Repair parts. - Repair parts shall be packed level A or B as specified in accordance with the applicable preservation and packaging specification (see 5. 2.1 1. 2). Unless otherwise specified in the contract or order, repair parts shall be packed separately and shipped concurrently with the basic equipment

5.2 1.3 Equipment manuals. - Service manuals shall be packaged level A or C and packed level A, B or C as specified in the contract or order in ac - cordance with MIL-M - 15071.

5.2.1.4 Draw ngs. - Drawin gs shall be packaged level A or C and packed level A, B or C as specified in the contract or order in accordance with MIL-D-963.

5.2.1.5 Marking - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Class required (see 1.2).
(c) Whether compressors shall include a supporting structure for sound isolation mounts (see 3.3.6.2).
(d) Whether electric lube oil heaters are required (see 3.3.15.7.1).
(e) Quantity of compressors per ship on which basis the quantity of onboard repair parts in a set shall be determined (see 3.6.1).
(f) Whether additional copies of manuals arc required (see 3.8. 3).
(g) Preservation, packaging or marking requirements other than those required by 5.1 (see 5.2).
(h) Whether repair parts boxes are required, if required specify type (metal or wood)
(i) Award will be made only for air compres - sors for which class C basic design drawings have been approved by the Bureau 01 Ships prior to the time set for opening of bids. Bidders should furnish with their bids a statement identifying by number the applicable, approved class C basic design drawings and identifying the letter or other action by the Bureau of Ships approving such draw - ings. (In the foregoing, for “bids” substitute “proposals” or “quotations” and for “'bidders” substitute “offerors” or “suppliers”, where appropriate.)

6.2 Prospective suppliers arc urged to submit to the Bureau of Ships, Code 649, class C basic design drawings of air compressors they propose to offer under this specification for Navy Shipboard use in order to become eligible for award of con - tracts and orders for such compressors

6.2.1 If a supplier, who has an approved class C basic design drawing for one or more of the dif - ferent classes of air compressors covered by this specification, desires to make any changes or revisions in his basic design for such class or classes of air compressors, he must submit new class C drawings or revision(s) to his previously approved class C basic design drawings, as appropriate, for approval. In the event a change or amendment to this specification is made, the Bureau of Ships will determine whether it is of such significance to basic

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designs as to warrant canceling any class C drawing approval(s) previously granted for any class or classes of compressors, If so, it will notify suppliers concerned in writing of such cancellation and such suppliers may, and are urged to, submit for approval new class C basic design drawings, or revisions to the formerly approved class C drawings, for the class or classes of compressors involved. If no notification of cancellation is given, any basic design approval granted prior to the change or amendment is automatically extended under this specification as changed or amended. ' Approval of new or revised class C drawings shall be effective only with respect to contracts and orders awarded subsequent to such approvals. Conditional approvals of class C basic design drawings or revisions thereto will not be made.

6.2.2 It is anticipated that the Bureau of Ships will require approximately 30 days after receipt of drawings to take approval action, but a longer time may be required if the drawings need revision or correction in order to be approved. This applies to each drawing submission, except that approval action on a relatively minor revision to a previously approved class C drawing may require less than 30 days. However, the Bureau of Ships cannot and does not undertake that it will take approval action in time to make a supplier eligible for any particular procurement or purchase of air compressors under this specification.

6.3 Table of compressor materials. -

Table I - Compressor materials

Application	Material	Applicable document 1/ 2/ 3/
Bedplate	Cast steel	MIL-S-15083, grade B or ASTM A27, grade 60-30
	or Structural steel	---
	or Forged steel	MIL-S-20166
	or Lead tin bronze	QQ-L-225 or ASTM A201, grade B
Cylinders	Valve bronze	MIL-B-16541 or ASTM B143, alloy 2A
	or Gun metal	MIL-M-16576 or ASTM B143, alloy 1A or 1B
	or Nodular (ductile) iron	MIL-I-17166 or ASTM A339, grade 60-45-10
Cylinder heads	Aluminum bronze	QQ-B-679 or ASTM B148
	or Aluminum manganese bronze	MIL-B-16522, class 2 or ASTM B147, alloy 8B
	or Valve bronze	MIL-B-16541 or ASTM B143, alloy 2A
	or Gun metal	MIL-M-16576 or ASTM B143, alloy 1A or 1B
Cylinder liners	Nitrided steel	MIL-S-869
	or Gray cast iron	QQ-I-652 or ASTM A48
	or Nitralloy G	Commercial
Pistons	Cold rolled, cold drawn steel	---
	or Cast aluminum	QQ-A-601 or ASTM B26
	or Gray cast iron	QQ-I-652 or ASTM A48
	or Forged steel	---
	or Stainless steel	---
Crankcases, frames and housings	Cast steel	MIL-S-15083, grade B or ASTM A27, grade 60-30
	or Structural steel	---
	or Nodular (ductile) iron	MIL-I-17166
	or Forged steel	MIL-S-20166
	or Lead tin bronze	QQ-L-[illegible]
Crankshaft	Forged steel	MIL-S-890, class An or B-S
	or Nodular (ductile) iron	MIL-I-17166

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Table I - Compressor materials (cont'd)

Application	Material	Applicable document 1/ 2/ 3/
Flywheel	Cast steel	MIL-S-15083, grade B
Connecting rods, piston rods, caps, bolts	Forged steel	MIL-S-890, class An or B-S
Wrist pins, crosshead pins	Forged steel or Nickel Molybdenum	MIL-S-890, class Ac Commercial
Housing tie rods	Forged steel	MIL-S-890, class Ac or An
Condensate separators:		
1500 p.s.i. pressure and below	Gun metal	MIL-M-16576 or ASTM B143, alloy 1A or 1B
	or Aluminum bronze	MIL-B-16033 or ASTM B148
above 1500 p.s.i. pressure	Steel tubing	MIL-T-20157, type E or ASTM A192
Water pump:		
Casing and impeller	Valve bronze	MIL-B-16541 or ASTM B143, alloy 2A
	or Gun metal	MIL-M-16576 or ASTM B143, alloy 1A or 1B
	or Nickel copper alloy	QQ-N-288
Shaft	Nickel copper alloy	QQ-N-281, class A or B or ASTM B164, class A or B
Heat exchangers (water through tubes):		
Shells	Valve bronze	MIL-B-16541 or ASTM B143, alloy 2A
	or Gun metal	MIL-M-16576 or ASTM B143, alloy 1A or 1B
	or Aluminum bronze	---
	or Naval brass	---
	or Seamless steel tubes	---
Tube sheets, heads and plates	Aluminum bronze	QQ-A-620
	or Aluminum bronze	QQ-A-630
	or Aluminum manganese bronze	MIL-B-16522
	or Copper nickel alloy	MIL-C-15762
Baffles	Copper nickel alloy	---
	or Brass	---
Tubes	Copper nickel alloy	MIL-T-15005 or MIL-T-16420 or ASTM B111
Heat exchangers (water surrounding tubes):		
Tubes	Copper nickel	MIL-T-15005 or MIL-T-16420 or ASTM B111
Tube sheets, heads and plates	Copper nickel alloy	MIL-C-15726
	or Valve bronze	MIL-B-16541 or ASTM B143, alloy 2A
	or Gun metal	MIL-M-16576 or ASTM B143, alloy 1A or 1B
	or Copper silicon alloy	MIL-C-17516
	or Aluminum bronze	QQ-A-620
	or Aluminum bronze	QQ-A-630
	or Aluminum manganese bronze	MIL-B-16522

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Table I - Compressor materials (cont'd)

Application	Material	Applicable document 1/ 2/ 3/
Heat exchangers (water surrounding tubes): (cont'd)		
Baffles	Copper nickel alloy or Brass	--- ---
Shells	Copper nickel alloy or Copper nickel alloy	MIL-T-15005 or ASTM B111 MIL-T-16420

1/ "ASTM" indicates standard specifications of the American Society for Test ing anti Mater ials (Application for copies should, be addressed to the American Society for Testing and Materials 1916 Race Street, Philadelphia 3, Pa.

2/ Materials used may be in accordance with either Government specifications or AsTM specifications, or equivalent material to any other recognized industry standard specification, such as Society of Automo - tive Engineers (SAE) and American Iron and Steel Institute (AISI), at the manufacturer's option

3/ Detail drawings and lists of material shall reference the specifications actually followed in each case, and shall include the class, type or grade of material used in each case, as applicable (see 3 4. 1 and 3.7.5.1).

Notice. - When Government drawings, specifi - cations, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever, and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or con~'eying any' rights or permission to manufacture, use, 01' sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

INSTRUCTIONS

BUDGET BU. NO. 45-R309

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

a. GIVE PARAGRAPH NUMBER AND WORDING

b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES" IN WHAT WAY?

4 REMARKS (Attach any pertinent data which may be of use in improving this specification. PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

SUBMITTED BY (Print name and activity)	DATE

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